

Chicago TNP Analytics and Event Business Simulation

Project Report
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Summary

This project builds an end-to-end analytics workflow for Chicago transportation network provider (TNP) trip data. The work starts with raw public trip files and ends with a business simulator for large events near Soldier Field. The full data set contains 243,479,296 completed trip records from 2022 through 2024. We used MatrixOne for large SQL operations and Python for modeling, validation, and simulation. This design keeps the full trip table in the database and moves only smaller summary data into memory.

The project has three connected goals. First, we organize and validate a very large trip data set. Second, we build models that describe and predict hourly demand, trip behavior, and event effects. Third, we turn the learned event pattern into repeated business scenarios. The final simulator asks practical questions such as: How many trips may appear during a game? When can the local fleet become too small? How do fixed, balanced, and aggressive pricing policies change service, rider spending, and contribution profit?

The final simulator is the main result of the current project. It learns normal demand, event demand, pickup and dropoff timing, route shares, fares, tips, and trip duration from real completed trips. It then generates 20,000 random two-day worlds. Each world contains an event day and a normal control day. The same generated demand and supply conditions are used for every fleet and pricing policy, so the policy comparison is fair.

Before the business simulation, the demand generator was tested on eight Bears games from 2024 that were not used to learn the event pattern. Pickup demand had a weighted absolute percentage error of 19.15%, and dropoff demand had an error of 22.78%. These results are not perfect, but they show that the generator can reproduce the main size and shape of a real event window. This validation is important because business results should not be calculated from an untested generator.

For the standard-fleet scenario, fixed pricing produced an average contribution profit of about \$59,432 across the two-day window. Balanced pricing increased the average to about \$65,885. Aggressive pricing produced about \$65,616. Balanced pricing had the highest average contribution profit, while aggressive pricing gave the highest original-demand fulfillment and a much lower severe-shortage probability. This is a useful example of a business trade-off: the policy with the strongest shortage protection is not exactly the policy with the highest average contribution profit.

All results must be read as scenario analysis. The source data contains completed trips, but it does not contain all ride requests, cancellations, real driver availability, or real platform accounting. Fleet size, price response, driver response, platform take rate, and cost are clear model assumptions. For this reason, the simulator supports comparison and planning; it does not claim to reproduce audited company profit or the exact Chicago driver market.

1. Project Background and Objectives

Large urban trip data can show strong patterns by time and place. Morning demand is different from evening demand. Weekends are different from weekdays. Airports, downtown areas, stadiums, and convention areas can have their own timing. A large event can also create a short dropoff peak before the event and a pickup peak after the event. A useful analytics system should find these patterns, test whether they repeat, and turn them into inputs for operations and planning.

The original project started with price and demand analysis. As the work developed, the focus became wider. We tested hourly H3 demand forecasts, pickup and dropoff relationships, spatial spread, recorded-tip prediction, large language model tasks, time-series foundation models, and cross-language deployment. We then selected a clear event case: Chicago Bears home games at Soldier Field. This case connects data engineering, demand prediction, event detection, and business simulation in one story.

The current project objectives are:

- Create a reliable full-data base from the 2022–2024 Chicago TNP files.
- Use H3 cells to compare demand across time and space.
- Predict short-term demand and test different model types on the same time split.
- Measure pickup and dropoff patterns around known events.
- Detect events when the schedule is not supplied.
- Generate realistic random demand scenarios and compare fleet and pricing policies.

The main design rule is a time-based test. Most models learn from 2022–2023 and are evaluated on 2024. We do not use a random row split for time-series work. A random split could allow future patterns to enter the training data and make the result look better than it would be in real use.

2. Data Source and Data Foundation

2.1 Source data

The data comes from the City of Chicago public TNP trip records. The project uses two full CSV files: one file for 2022 and one file covering 2023–2024. Together they contain 243,479,296 completed trip records. The 2022 source contributes 69,109,780 rows after loading, and the 2023–2024 source contributes 174,369,516 rows. The files include trip time, trip seconds, trip miles, fare, tip, additional charges, trip total, shared-trip fields, community areas, and pickup and dropoff centroid locations.

The files do not have exactly the same schema. The later source contains three fields that do not exist in the 2022 file. The timestamp text also uses a different format. We first load each source into its own all-text staging table. We then convert the fields into a common typed table. Missing source-only fields receive null values. Timestamps become database datetime values, and numeric columns become numeric types. This staging-first design makes large imports safer because source problems can be checked before type conversion.

2.2 MatrixOne and H3

MatrixOne is the main storage and aggregation layer. The full trip data stays in MatrixOne. SQL is used for row counts, schema checks, time filters, joins, hourly aggregation, and event data extraction. Python receives only the smaller data needed by a model or chart. This is important because loading all 243 million rows into a local pandas DataFrame would require too much memory and would make the workflow hard to repeat.

The project converts pickup and dropoff centroid locations into H3 resolution-9 cells. H3 changes a map location into a standard cell identifier. The source contains 879 unique public centroids, which map to 858 distinct H3 cells. Some H3 cells near Soldier Field do not have trip values. This does not mean that nobody traveled there. The public data reports a limited set of centroid locations instead of exact passenger coordinates. For event work, we therefore select nearby H3 cells that have real observed activity, rather than selecting cells only by geometric distance to the stadium.

Item	Validated value
Full clean and H3 table	243,479,296 trip rows
2022 source	69,109,780 rows
2023–2024 source	174,369,516 rows
Unique public centroids	879
Distinct H3 resolution-9 cells	858
Main model split	Train on 2022–2023; test on 2024

3. Main Analytical Work

The project notebooks follow one main line. Early notebooks prepare and test the data. Middle notebooks compare forecasting and machine learning methods. Later notebooks apply the methods to a real event and then build the final simulator. The following summary explains the role of each stage.

Stage	Main work	Main output
01	Data ingestion, validation, cleaning, and H3 preparation	Validated clean and H3 tables
02	Hourly H3 demand and spatial dynamics	Short-term forecasts and flow signals
03	Two-stage recorded-tip prediction and LLM/tree test	Tip amount and yes/no benchmarks
04	LLM peak-continuation benchmark	1-, 3-, and 6-hour peak tests
05	Chronos-2 and LightGBM comparison	Aligned 24-hour forecasts

Sta ge	Main work	Main output
06	ONNX and Go deployment validation	Portable next-hour model
07	Bears event sensitivity and blind detection	Known-event prediction and blind Top-K results
08	Event business Monte Carlo simulator	20,000 business scenarios

3.1 Hourly H3 demand and spatial flow

The hourly demand work studies four related questions. It measures how far an actual hourly pickup count is from the normal count for the same H3, weekday, and hour. It predicts pickup demand one, two, three, and six hours ahead. It tests whether dropoffs lead later pickups. It also checks whether demand growth spreads to nearby H3 cells. Lag and rolling features use only earlier hours, which prevents future information from entering a prediction.

The selected one-hour H3 model reaches an MAE of 12.66 pickups and an R-squared value of 0.9412 on the 2024 test period. MAE means the average absolute difference between the prediction and the observed count. R-squared measures how much of the change in demand is explained by the model. The result shows that the model captures most normal hourly movement in high-activity H3 cells. Longer forecasts are harder because uncertainty grows with each added hour.

The anomaly part separates the current level from the direction of change. A location can remain above its normal baseline while demand is falling quickly from the previous hour. For this reason, the project uses both the current z-score and the change in z-score. Pickup and dropoff are also studied together. A dropoff peak can add vehicles to an area, and the same area can later show a pickup peak. This relationship becomes an important input to the final supply simulation.

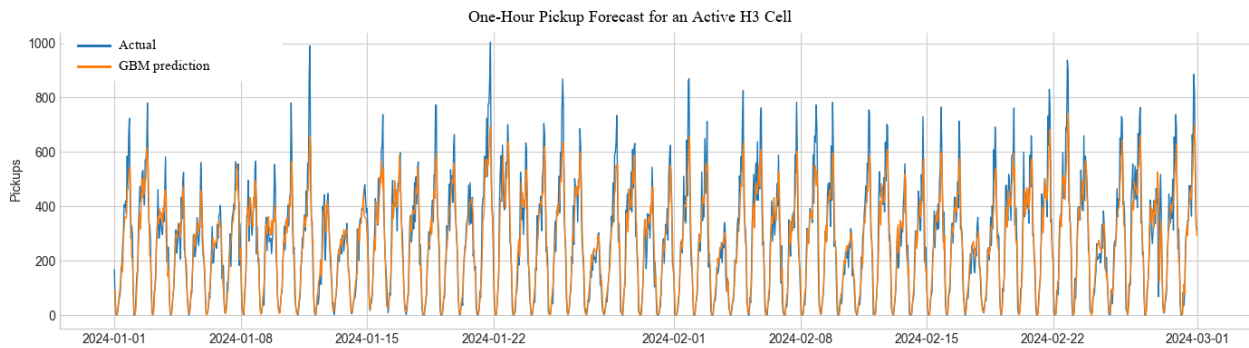


Figure 1. Example of actual and predicted one-hour pickup demand for one active H3 cell in March 2024.

The blue line is the observed hourly pickup count and the orange line is the model prediction. The model follows the repeated daily rise and fall and also follows many short peaks. The remaining gaps show why the MAE is not zero: unusual local spikes are harder to predict than the normal daily cycle.

3.2 Recorded-tip prediction

The recorded-tip model uses a two-stage structure. Stage one predicts whether the data records a positive tip. Stage two predicts the amount only for trips with a positive tip. The final expected tip is the probability from stage one multiplied by the positive amount from stage two. This structure is more suitable than one direct model because the data contains many zero-tip rows and a smaller group of positive amounts.

The training-mean baseline has an expected-tip MAE of about \$1.576. Models using pickup time and location reduce the MAE to about \$1.415. Adding actual dropoff time and location change produces little extra improvement in this sample. The result suggests that trip-level tip behavior is difficult to predict precisely from location and time alone. It also shows why the target is called recorded tip: cash tips that are not recorded by the platform are not visible in this data.

A second tip experiment compares an OpenAI model with tree models on the yes/no question. All methods use the same 200 test trips and do not receive the true tip field. The sample has a strong no-tip majority. A shallow tree reaches high overall accuracy mainly because it predicts no tip for most rows. The LLM finds more positive-tip cases but has lower total accuracy. This test shows that accuracy alone can be misleading when one class is much more common. Recall, balanced accuracy, and F1 are needed for a fair view.

3.3 LLM and time-series model tests

The LLM peak-continuation test asks a narrow question: after the current hour is confirmed as a peak, will the peak still be active one, three, or six hours later? The LLM does not receive millions of rows. It receives one small structured summary with recent demand, baseline values, z-scores, lag values, and related evidence. This is the practical way to connect a large database to an LLM. The database prepares small cases, and the LLM makes a decision or explanation for each case.

The 60-case feasibility test shows that the best method depends on the forecast horizon. A simple current-hour rule is competitive at one hour. A gradient-boosting model performs best at three and six hours in the recorded run. The LLM produces valid structured answers and readable reasons, but it does not beat the strongest traditional method on numeric accuracy. This result supports a mixed design: use traditional models for numbers and use an LLM for explanation, review, or external event search.

The Chronos-2 benchmark compares a time-series foundation model with LightGBM on the same 24-hour forecast windows. It uses 121 forecast origins and 29,040 prediction rows. Each origin uses 28 days of history to predict the next 24 hours. In the recorded run, Chronos-2 with history only has the best overall MAE, about 30.41. This value is not directly comparable with the one-hour MAE of 12.66 because the models use different horizons, H3 groups, and evaluation rows.

3.4 Portable model deployment

The selected next-hour H3 model was exported to ONNX. ONNX stores the trained model in a language-neutral format. A feature schema fixes the order of 61 inputs, the training medians, valid H3 codes, tensor names, and non-negative post-processing. Python ONNX Runtime and a Go program read the same model and the same test cases. A parity test requires their predictions to

match within 0.001 pickup. Passing this test shows that a model trained in Python can run in a Go service without changing its result.

Test	Main target	Evaluation design	Recorded result
Hourly H3 model	Next-hour pickups	2022–2023 train; 2024 test	MAE 12.66; R ² 0.9412
Recorded-tip model	Expected recorded tip	Time-based holdout	MAE reduced from \$1.576 to about \$1.415
LLM peak test	Peak continues after 1/3/6 hours	60 structured cases	Feasible outputs; tree/GBM stronger on numeric accuracy
Chronos-2	Next 24 hourly values	121 origins; 29,040 rows	Best recorded MAE about 30.41
ONNX and Go	Same next-hour prediction	Python versus Go parity	Difference required below 0.001 pickup

4. Bears Event Analysis and 2024 Test

The event study uses Chicago Bears home games at Soldier Field as a clear and repeatable test case. The model learns from 17 home games in 2022–2023. For every game, the data is aligned to kickoff time. Negative relative hours are before kickoff, and positive relative hours are after kickoff. The model compares each event hour with normal non-game hours that have a similar weekday, clock hour, and nearby date.

The historical event curve has a clear direction. Dropoffs increase before the game as people arrive. Pickup demand rises later as people leave. In the learned zone-level template used by the simulator, the average dropoff effect reaches about 4.62 times normal around two hours before kickoff. The pickup effect reaches about 2.42 times normal around three hours after kickoff. Individual H3 cells can have larger or smaller effects, which helps the model place event traffic across the selected area.

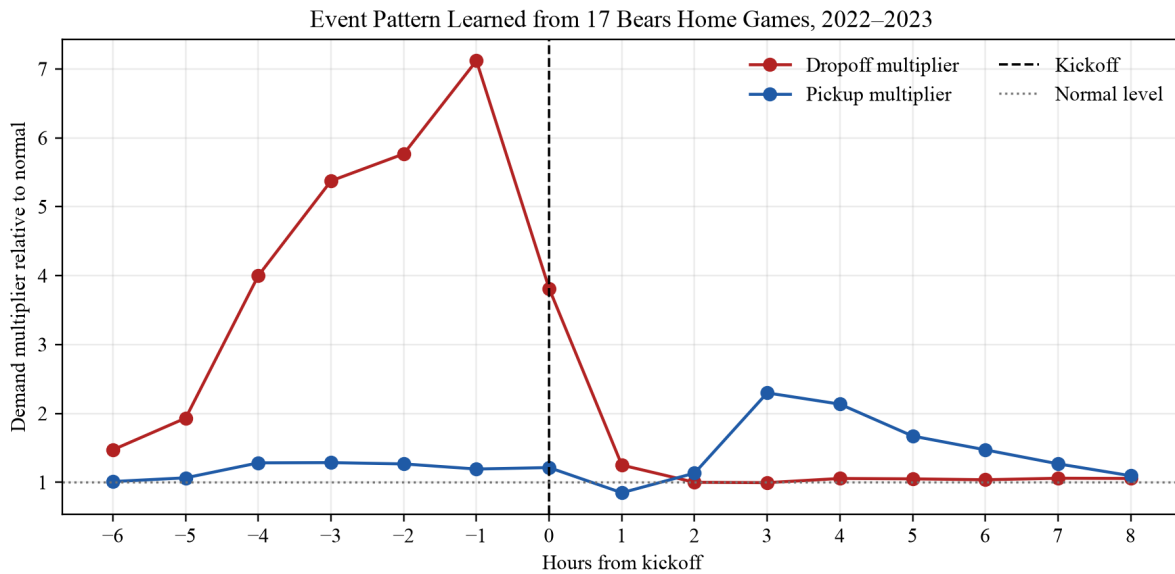


Figure 2. Pickup and dropoff demand multipliers learned from 17 Bears home games in 2022–2023.

The horizontal normal level is 1.0. Values above 1.0 mean more trips than a similar non-game hour. The red dropoff curve rises well before kickoff and reaches its highest point around one to two hours before the game. The blue pickup curve becomes strongest several hours after kickoff. The order of these two peaks matches the expected movement of spectators into and out of the area.

The 2024 work has two tests. In the first test, the model receives the real 2024 game date and kickoff time. It predicts every relative hour using the event template learned from 2022–2023. Against a normal-day baseline, pickup MAE improves by 44.4%, and dropoff MAE improves by 68.5%. This shows that the learned event shape adds useful information when the event schedule is known.

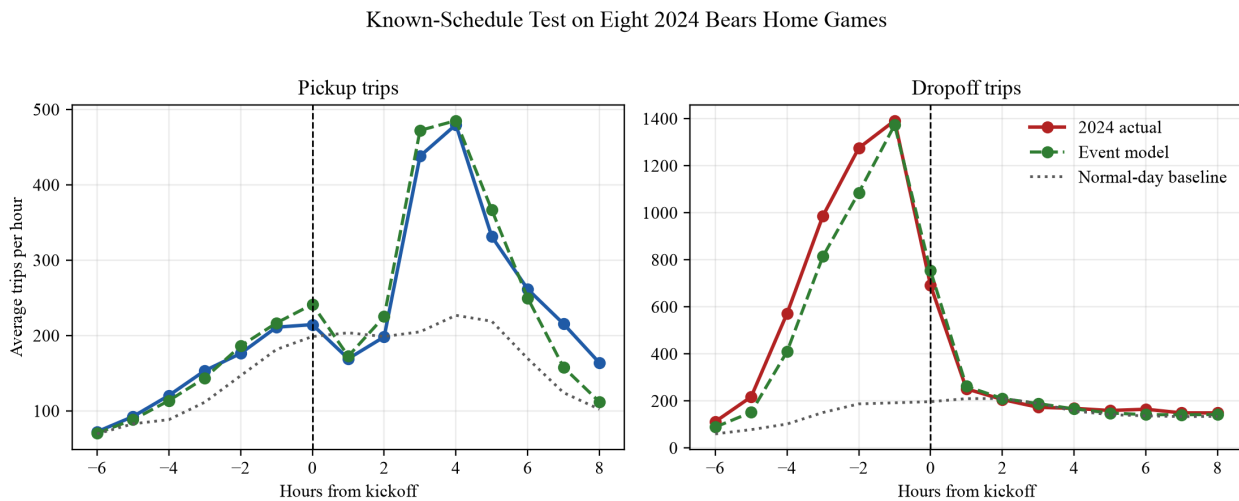


Figure 3. Average hourly prediction for the eight 2024 games when the game schedule is known.

The colored solid line is the 2024 observed average, the green dashed line is the event-model prediction, and the gray dotted line is the normal-day baseline. The event model is much closer to

the actual pre-game dropoff rise and post-game pickup rise than the normal baseline. This explains the large MAE improvement, especially for dropoffs.

Flow	Event-model MAE	Normal baseline MAE	WAPE	MAE improvement
Pickup	41.43	74.47	18.84%	44.37%
Dropoff	95.50	303.17	21.49%	68.50%

In the second test, the model does not receive the 2024 schedule. It scans candidate times and ranks them by event-like features. After the candidate list is fixed, it is compared with the real schedule. Five of the eight true games appear in the Top 8 candidates. All eight appear in the Top 25 candidates, but precision falls because more non-game candidates are included. This is a useful screening system, not a final event calendar. It can reduce a full year to a smaller list for review or for an LLM-based event search.

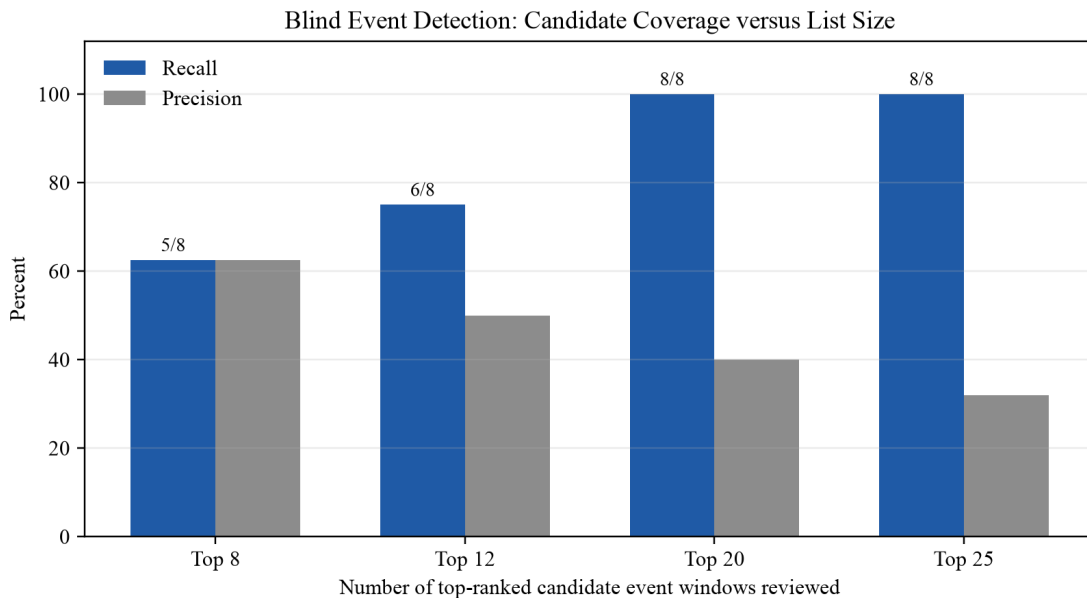


Figure 4. Blind event detection coverage as the candidate review list becomes longer.

Recall measures how many of the eight real games are found. Precision measures how many reviewed candidates are true games. A short Top 8 list is easier to review and finds five games. A Top 20 or Top 25 list finds all eight games, but it also includes more false candidates. This is the normal trade-off between missing events and reviewing extra alerts.

Candidate list	Games found	Recall within 2 hours	Precision within 2 hours
Top 8	5 of 8	62.5%	62.5%
Top 12	6 of 8	75.0%	50.0%
Top 20	8 of 8	100.0%	40.0%
Top 25	8 of 8	100.0%	32.0%

5. Event Business Monte Carlo Simulation

5.1 Why the simulator is needed

A normal forecast gives one expected demand curve. It can answer a question such as, 'How many pickups may happen after this game?' Business planning needs more than one expected curve. Actual events vary. Demand can be higher or lower than the average. Vehicles may arrive from dropoffs at different rates. Some idle vehicles may leave the small area. Price changes can reduce rider demand and may attract additional supply. A single forecast cannot show the full range of these outcomes.

The simulator adds controlled randomness around patterns learned from real data. It creates many possible worlds rather than one fixed answer. We can then report the average result, a low result, a high result, and the chance of a shortage. We can also apply several business policies to exactly the same generated world. This makes the simulator useful for stress testing. The question changes from 'What is the forecast?' to 'Across many reasonable outcomes, how often does this policy work, and what risks remain?'

5.2 Geographic scope and H3 selection

The simulation uses six active H3 resolution-9 cells around Soldier Field. Two cells are prevalidated core cells from the event study. Four additional cells are selected from nearby cells with real pickup or dropoff activity. A cell must have at least 1,000 historical trips and must be within two kilometers of the stadium. Geometric neighbor cells are still shown on the map, but empty geometry-only cells are not used to generate demand.

This selection rule handles an important data limit. The stadium marker can appear inside an H3 cell with no public TNP centroid. The closest visible H3 cell is therefore not always the closest cell with reported trip activity. The selected cells represent where the public data places passengers around the event area. The model does not claim that every pickup happened at the center of each H3 cell.

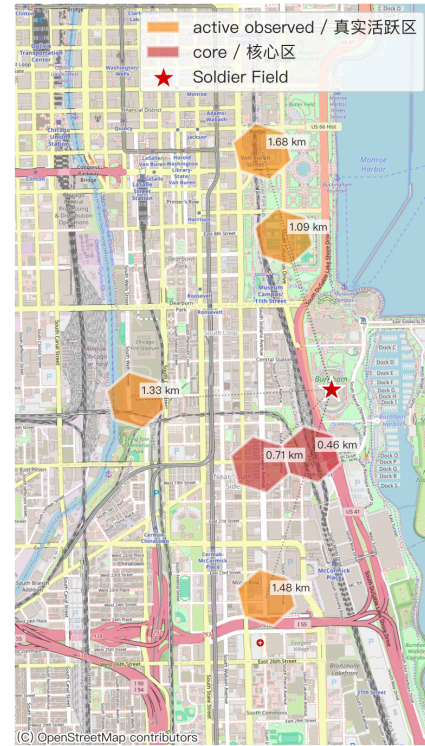
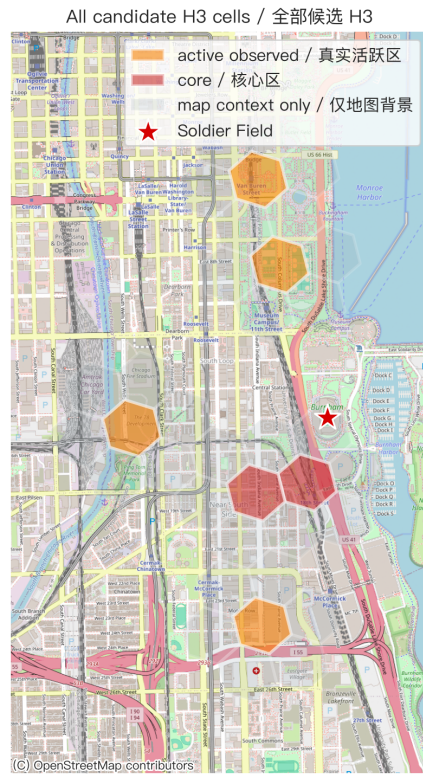


Figure 5. Soldier Field, all nearby H3 context cells, and the six active observed cells used by the simulator.

The red star marks Soldier Field. The red cells are the two core cells carried forward from the event study. The orange cells are additional nearby cells with enough observed TNP activity. Gray outlines provide map context only. This map makes clear why the selected cells are not simply the six geometric cells closest to the stadium: the public source records trips at a limited set of centroids, so the model uses the closest cells that contain usable activity.

5.3 What the simulator learns from real data

The simulator learns five main groups of information from MatrixOne:

- Normal hourly pickup and dropoff demand for each selected H3 cell, weekday, and clock hour.
- The event multiplier by H3 cell and relative hour around kickoff.
- The normal size and time correlation of demand errors, so adjacent hours move in a realistic way.
- Pickup-to-dropoff route shares, including average distance, duration, and fare for each route.
- Recorded tip rate and positive-tip amount by pickup H3 and clock hour.

The event effect is not a manual curve. It is estimated from the 17 training games. The model first calculates the normal demand for each game hour. It then measures the observed event demand above that normal value. The average event effect is stored by relative hour and H3 cell. Random event strength is limited to a range close to historical games, which prevents the generator from creating unrealistic extreme events.

Event Effect Used by the Demand Generator

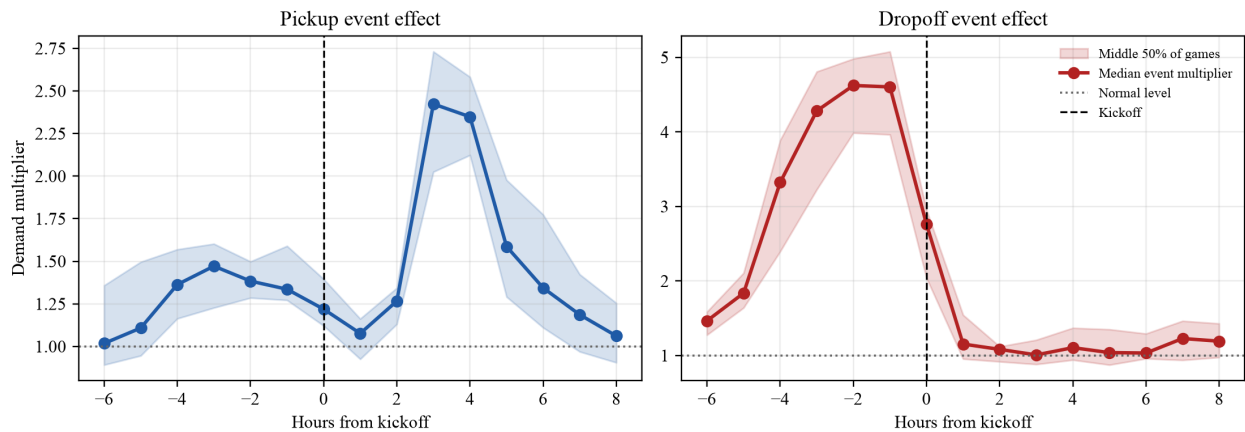


Figure 6. Median event multiplier and the middle 50% range learned from the training games.

The center line is the median event effect and the shaded band shows the middle half of the training games. The band is important because games do not create exactly the same demand. Dropoff demand has the strongest effect before kickoff, while pickup demand peaks after the game. The simulator samples within this learned variation instead of repeating one fixed curve.

5.4 Independent demand validation

The generator is checked on the eight 2024 games before it is used for pricing or profit analysis. These games are outside the 2022–2023 learning period. For each game and relative hour, the generator creates a prediction interval and a median prediction. The result is compared with the actual pickup and dropoff counts.

Flow	Actual total	Predicted total	MAE	WAPE	90% interval coverage
Pickup	66,018	66,883	105.38	19.15%	83.33%
Dropoff	84,715	84,167	160.78	22.78%	82.50%

The predicted totals are close to the actual totals, while the hourly WAPE values show that the exact timing and cell allocation still have error. The 90% interval covers about 83% of the actual hourly points, which means the interval is somewhat narrower than a fully calibrated 90% interval. These results are strong enough for a planning prototype, but they also show where more event data and better supply data can improve the model.

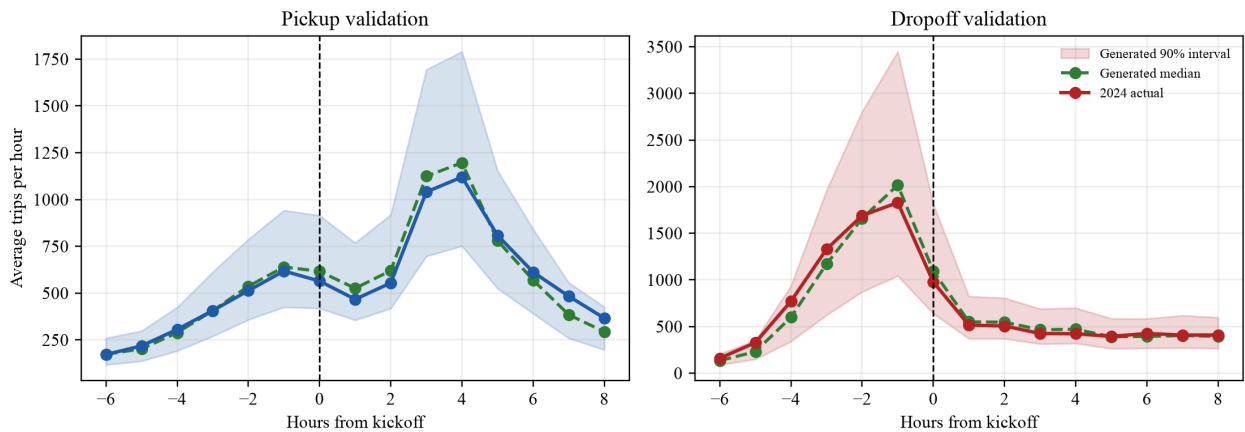


Figure 7. Generated demand interval, generated median, and actual demand for eight unseen 2024 games.

The blue and red solid lines are the actual average pickup and dropoff counts. The green dashed line is the generated median. The shaded area is the generated 90% interval. Most actual points follow the same event shape and fall inside the range, but the interval misses some large hourly values. This supports use as a planning model while also showing that the uncertainty band should be recalibrated when more event data becomes available.

5.5 How one random world is generated

Each simulation covers 48 hours. The first day contains a game at the configured kickoff time. The second day is a control day without the game effect. For each hour and H3 cell, the generator starts with normal demand. On the event day, it applies the learned event multiplier. It then adds random changes based on the historical error distribution. The random changes are linked across nearby hours, so the generated curve does not jump independently from one hour to the next.

The generator separately creates pickup requests and dropoff arrivals. It also samples an event-strength value, a dropoff-to-available-vehicle rate, an idle-vehicle retention rate, and an initial fleet scale. These random values are shared by every strategy inside the same world. This shared-world design is important. If two pricing policies used different random demand, the difference in profit could come from different luck rather than from the policy.

The simulator first works with H3-by-hour counts because this is much faster than creating every trip before the business test. After the 20,000 runs are complete, it selects representative low, median, and high profit runs. It then creates full synthetic request records for these runs. Each record has a request time, pickup H3, sampled dropoff H3, trip miles, trip seconds, base fare, recorded tip, price multiplier, acceptance flag, served flag, and rider payment. Route, distance, duration, fare, and tip distributions come from the real training data. The records are synthetic and are marked as synthetic data.

5.6 Fleet scenarios

The reference fleet is based on the 95th percentile of normal hourly pickups in each selected H3 cell. This is a planning reference, not the observed number of real Chicago drivers. Four fleet

factors create different stress levels. The simulator also allows only 30% of vehicles above a two-vehicle local reserve to move from a surplus selected cell to a shortage selected cell during one hour. This prevents unrealistic instant movement of the full fleet.

Fleet scenario	Factor	Simple meaning
Constrained	35%	Strong shortage stress; much less than the reference fleet
Lean	65%	Limited fleet with frequent pressure
Standard	100%	Reference planning fleet
Ample	135%	More vehicles than the reference fleet

5.7 Pricing strategies

Pricing reacts to pressure, where pressure equals base pickup requests divided by available vehicles in the H3 cell. No policy increases price while pressure is 0.80 or lower. Above 0.80, the balanced and aggressive policies increase the multiplier at different speeds. The multiplier then affects retained rider demand and can add nearby supply. The fixed strategy never changes price.

Policy	Price rule	Demand response	Supply response	Driver incentive
Fixed	Always 1.00x	No price suppression	No extra supply	None
Balanced	$1 + 0.35 \times \text{pressure above } 0.80$; cap 1.50x	Elasticity - 0.20	0.25 response	12% of added surge fare
Aggressive	$1 + 0.65 \times \text{pressure above } 0.80$; cap 2.00x	Elasticity - 0.35	0.45 response	20% of added surge fare

The names describe the strength of the response. Balanced pricing has a smaller price coefficient, a lower maximum multiplier, a smaller demand reduction, and a smaller supply response. Aggressive pricing reacts faster, can reach 2.00x, removes more price-sensitive requests, and assumes a stronger supply response. These are clear scenario inputs. They are not estimates of the real price algorithm of any company.

5.8 Hour-by-hour operating logic

At the start of each simulated hour, some idle vehicles remain in the local area and some leave. Completed dropoffs add possible vehicles, but only the sampled available share is counted. The pricing policy reads the pressure and calculates a price multiplier. The demand-elasticity assumption converts base requests into retained requests. Dynamic pricing may also attract a limited amount of extra nearby supply.

The simulator then moves a limited share of surplus vehicles between the six selected cells. Completed trips equal the smaller of retained requests and available vehicles. Retained requests without a vehicle become unserved trips. Base requests that leave because of price become price-

suppressed requests. These two groups are kept separate. A policy should not receive service credit simply because a higher price removed a request.

Rider spending equals charged fare plus recorded tip for completed trips. Platform revenue equals charged fare multiplied by the assumed 25% take rate. Driver incentive cost equals the added surge fare multiplied by the policy incentive share. Variable operating cost is \$0.35 per completed trip. Contribution profit equals platform revenue minus driver incentive cost and variable operating cost. It does not include fixed company costs, taxes, insurance, support, marketing, or other accounting items.

5.9 The 20,000-run experiment

The full experiment creates 20,000 random two-day worlds. Each world is tested with four fleet sizes and three pricing policies, which gives 12 policy combinations per world. The simulator writes checkpoint files in batches, so a stopped run can continue without repeating completed worlds. Final validation checks that all 20,000 run identifiers are present, all 12 combinations exist, counts are non-negative, request accounting balances, service rates stay between zero and one, and representative trip files were created.

The result is a distribution rather than one number. P05 is a low outcome: only about 5% of generated worlds are below it. P50 is the median. P95 is a high outcome: only about 5% of worlds are above it. This range shows uncertainty that a single forecast cannot show. It also supports policy comparisons under both normal and difficult random conditions.

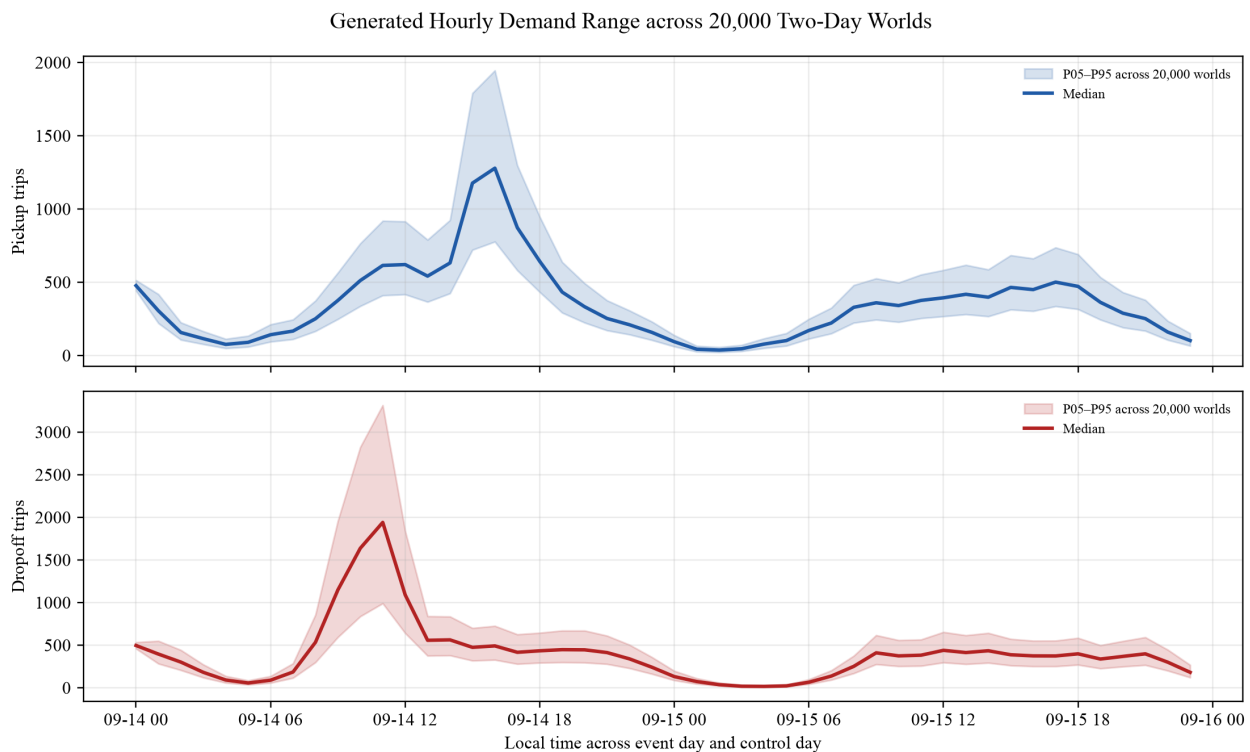


Figure 8. Hourly pickup and dropoff demand ranges generated across 20,000 two-day worlds.

The dark lines show the median generated demand and the shaded areas show the P05–P95 range. The first day contains the event and the second day is the control day. The pickup range expands

after the event and the dropoff range expands before it. This is the main benefit of simulation: planning can use a range of reasonable outcomes instead of relying on one average curve.

5.10 Main business metrics

The simulator reports several related metrics. They should be read together:

- Base requests: generated rider demand before any price response.
- Retained requests: requests that remain after the assumed response to price.
- Completed trips: retained requests that receive a vehicle.
- Price-suppressed requests: base requests that leave after a higher price.
- Unserved trips: retained requests that cannot receive a vehicle.
- Operational service rate: completed trips divided by retained requests.
- Original-demand fulfillment: completed trips divided by base requests.
- Stockout probability: share of worlds with at least one unserved retained request.
- Severe-shortage probability: share of worlds where unserved requests are at least 5% of base demand.
- Contribution profit: assumed platform revenue minus incentives and per-trip variable cost.

5.11 Standard-fleet results

The standard fleet is the clearest central comparison. The table below covers the full 48-hour event and control window. All three policies see the same 20,000 random worlds.

Policy	Completed trips	Original-demand fulfillment	Severe-shortage probability	Mean price	Mean contribution profit
Fixed	14,278	82.65%	95.84%	1.00x	\$59,432
Balanced	14,379	83.22%	90.57%	1.18x	\$65,885
Aggressive	14,681	84.92%	52.42%	1.32x	\$65,616

Balanced pricing increases average contribution profit by about \$6,453 compared with fixed pricing. It completes about 101 more trips, reduces unserved demand by about 765 requests, and suppresses about 665 requests through price. Aggressive pricing increases average contribution profit by about \$6,184. It completes about 402 more trips, reduces unserved demand by about 2,036 requests, and suppresses about 1,634 requests through price.

Aggressive pricing has the highest original-demand fulfillment because the assumed supply response is stronger and fewer retained requests remain unserved. At the same time, riders spend about \$95,013 more than under fixed pricing across the two-day window. Balanced pricing increases rider spending by about \$48,205. This shows why profit, fulfillment, shortage risk, and rider spending must be reviewed together.

The stockout probability remains high for all standard-fleet policies because the metric becomes true after even one unserved retained request. Severe-shortage probability is more useful for

comparing material stress. Fixed pricing has a severe-shortage probability of 95.84%, balanced pricing has 90.57%, and aggressive pricing has 52.42%. The values depend strongly on the fleet and supply assumptions; they should not be presented as observed Chicago shortage rates.

Across fleet sizes, the result changes in a reasonable direction. Constrained and lean fleets show more lost demand and stronger shortage. The ample fleet completes a much larger share of original demand and needs less price pressure. This behavior is an important model check. A simulator would not be useful if fleet size had no clear effect on service.

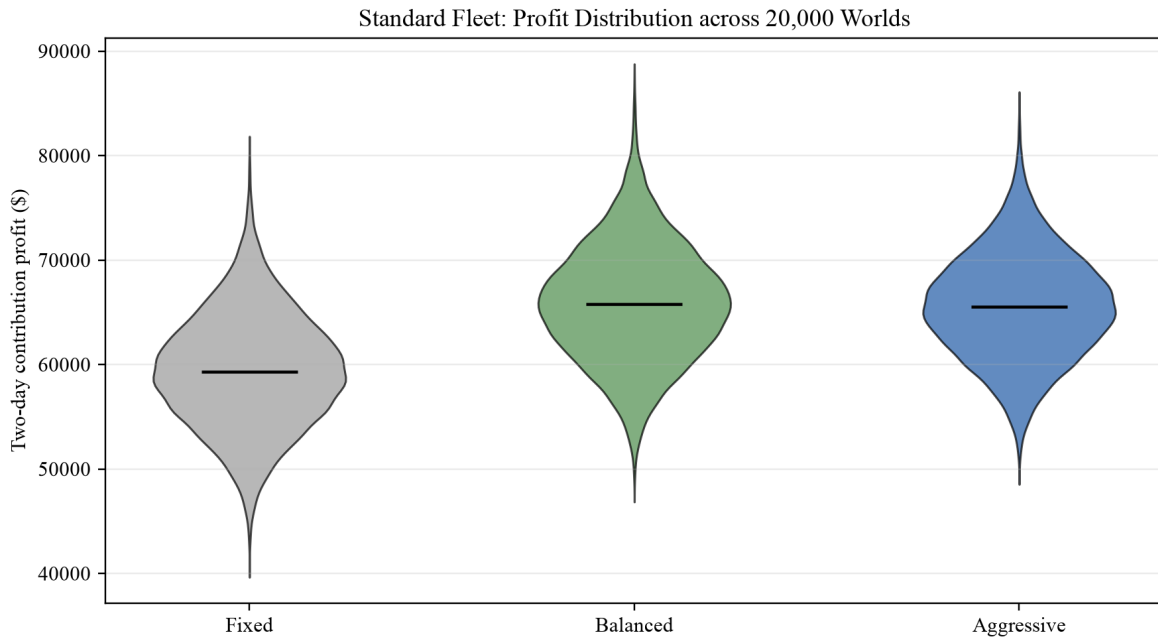


Figure 9. Distribution of two-day contribution profit for the standard fleet under the three pricing policies.

Each shape contains 20,000 profit outcomes. Wider sections mean that more simulated worlds fall in that profit range, and the black line marks the median. Balanced and aggressive pricing both shift the distribution above fixed pricing. Balanced has a slightly higher center than aggressive, while the ranges still overlap. This is why the conclusion should be based on the full distribution and not only one average value.

Fleet Capacity and Pricing Policy Trade-offs

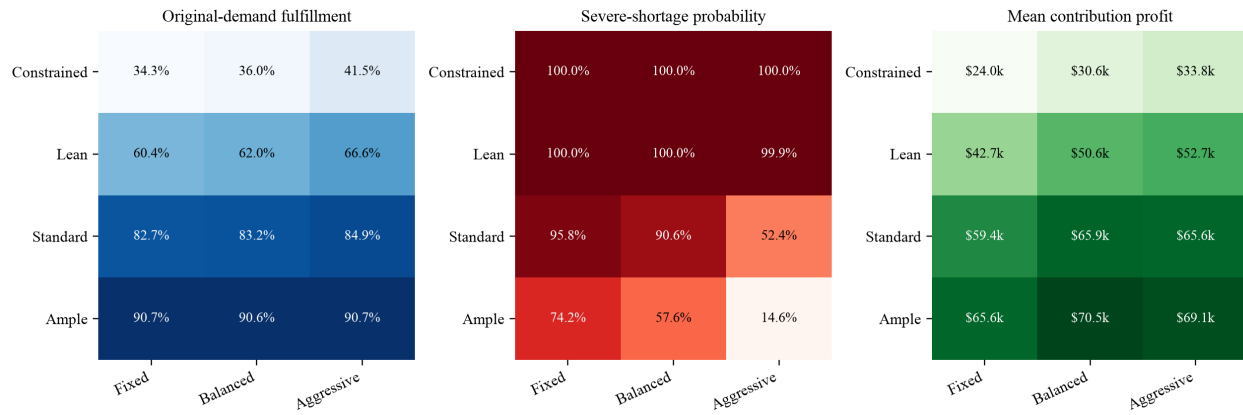


Figure 10. Original-demand fulfillment, severe-shortage probability, and mean contribution profit for all fleet and policy combinations.

Each row is a fleet level and each column is a pricing policy. Darker blue means higher original-demand fulfillment, darker red means higher severe-shortage risk, and darker green means higher mean contribution profit. More fleet capacity improves service much more than pricing alone. Aggressive pricing produces the largest shortage reduction for the standard and ample fleets, while balanced pricing gives the highest mean contribution profit for those two fleet levels.

Fleet	Policy	Original-demand fulfillment	Severe shortage	Mean price	Mean contribution profit
Constrained	Fixed	34.34%	100.00%	1.00x	\$24,004
Constrained	Balanced	36.02%	100.00%	1.37x	\$30,563
Constrained	Aggressive	41.52%	100.00%	1.75x	\$33,815
Lean	Fixed	60.43%	100.00%	1.00x	\$42,709
Lean	Balanced	61.96%	100.00%	1.27x	\$50,551
Lean	Aggressive	66.61%	99.89%	1.52x	\$52,749
Standard	Fixed	82.65%	95.84%	1.00x	\$59,432
Standard	Balanced	83.22%	90.57%	1.18x	\$65,885
Standard	Aggressive	84.92%	52.42%	1.32x	\$65,616
Ample	Fixed	90.65%	74.17%	1.00x	\$65,573
Ample	Balanced	90.61%	57.63%	1.13x	\$70,549
Ample	Aggressive	90.71%	14.59%	1.23x	\$69,139

5.12 Business Interpretation and Scenario Questions

The current simulator supports controlled business questions by changing the event date and event strength, fleet factor, price cap, price sensitivity, supply response, take rate, incentive share, or variable cost. The same number of random worlds can then be evaluated under each policy. This makes the analysis a policy stress test rather than only a report on one historical event.

The main scenario questions are:

- How much local fleet is needed to keep severe-shortage risk below a selected level?
- Does a higher price cap improve service enough to justify higher rider spending?
- How much driver incentive can be paid before contribution profit falls below fixed pricing?
- How does a stronger or weaker event change the range of completed trips and profit?
- Which H3 cells need early supply before kickoff, and which cells need supply after the event?

A new policy can be added through the explicit strategy fields without changing the demand generator. This separation keeps the comparison clear: the same random worlds test different operating rules, so differences in the results come from the policy rather than from different random demand.

6. Deliverables and Reproducibility

The project is organized as a sequence of readable notebooks. The database connection, project directory, and API keys are provided through environment variables or hidden runtime input. Source notebooks do not store local passwords or API keys. Large intermediate results use named cache and output folders. This allows a long query or model run to be reused without rebuilding the full data set every time.

The main technical deliverables include the validated MatrixOne clean and H3 tables, hourly demand caches, model metrics, tree rules, LLM response caches, Chronos and LightGBM predictions, the ONNX model package, Go parity tests, Bears event curves, blind event candidates, demand-generator validation files, Monte Carlo summary files, policy comparison tables, and representative synthetic trip exports.

The final simulator also exports a real map with the stadium, selected H3 cells, and context cells. It exports the learned event template, H3-level event sensitivity, demand intervals, business quantiles, incremental policy results, and selected synthetic orders. These files support review at summary level and inspection of selected trip-level examples.

7. Important Limits

The first limit is the definition of demand. The public TNP data contains completed trips. It does not include every quote, ride request, failed match, cancellation, wait time, or available driver. A low completed-trip count can mean lower rider demand, lower supply, or a data issue. The model cannot fully separate these causes without more marketplace data.

The second limit is location precision. H3 cells are created from published centroids, not exact passenger coordinates. Empty cells around the stadium can still be part of the real walking and

pickup area. The selected six cells are the nearest useful observed cells under the stated rules. Exact or finer pickup points would improve the spatial model.

The third limit is the number of event examples. The event template learns from 17 Bears home games and is tested on eight games. This is enough for a clear proof of concept, but it is not enough to describe every type of event. Concerts, conventions, bad weather, holiday events, and events at other venues can have different arrival and exit patterns.

The fourth limit is the business assumptions. Real fleet size, driver acceptance, driver movement, price elasticity, supply response, platform take rate, incentives, and cost are not present in the public trip data. The simulator keeps these values visible so that they can be replaced with operating estimates. Until real values are supplied, the profit and shortage results are planning scenarios, not audited forecasts.

8. Next Development Priorities

8.1 Calibrate with marketplace data

The highest-value next step is to replace scenario assumptions with operating data. Useful fields include request count, quote count, accepted trips, canceled trips, driver online count, available driver count, wait time, pickup ETA, driver movement, price multiplier, and actual incentive cost. These fields would allow the simulator to estimate real demand loss, real supply response, and real service risk.

8.2 Expand event types and external features

The event framework can be expanded beyond Bears games. A larger event library can include concerts, conventions, baseball games, festivals, airport disruptions, holidays, and weather. Each event type can have its own arrival and exit curve. Weather and event information should be joined as stable local files or database tables, rather than downloaded during every notebook run.

8.3 Build a policy review interface

A simple review interface can allow users to choose the event, event strength, fleet size, pricing rule, incentive share, and cost assumptions. The interface can show P05, median, mean, and P95 results for completed trips, shortage, rider spending, and contribution profit. It should also show the H3-hour demand and supply curve so that policy effects can be traced by time and location.

9. Conclusion

This project shows a complete path from very large public trip files to a practical event-planning simulator. MatrixOne supports the full-data layer, H3 gives a common spatial unit, forecasting models describe short-term demand, event analysis finds repeatable pickup and dropoff patterns, and the Monte Carlo simulator turns those patterns into business ranges. The strongest value is not one final profit number. The value is a repeatable system that can compare decisions under uncertainty and can become more accurate when real marketplace data is added.